Euler's Pentagonal Number Theorem

Formalization via Franklin's involution & the Jacobi triple product

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The theorem

Euler's Pentagonal Number Theorem

$$\prod_{n=1}^{\infty}(1-x^n) \;=\; \sum_{k=-\infty}^{\infty}(-1)^k\,x^{k(3k-1)/2} \;=\; 1+\sum_{k=1}^{\infty}(-1)^k\left(x^{\frac{k(3k-1)}{2}}+x^{\frac{k(3k+1)}{2}}\right).$$

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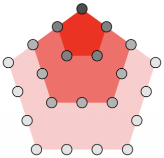
Combinatorial form. Let $p_e(n)$ (resp. $p_o(n)$) count partitions of n into an *even* (resp. *odd*) number of *distinct* parts. Then

$$p_e(n)-p_o(n)=\begin{cases}(-1)^k & n=\frac{3k^2\pm k}{2} \quad \text{(pentagonal)},\\ 0 & \text{otherwise.}\end{cases}$$



k	-3	-2	-1	0	1	2	3
$\frac{3k^2-k}{2}$	15	7	2	0	1	5	12

Generalized pentagonal numbers $\frac{3k^2-k}{2}$ ($k \in \mathbb{Z}$): $0, 1, 2, 5, 7, 12, 15, \ldots$

Base, slope, and Franklin's involution

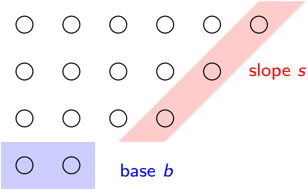


Diagram of $17 = 6 + 5 + 4 + 2$.

$b = \min(S)$, $m = \max(S)$,
$s =$ length of the top consecutive run,
$D = \{m - s + 1, \dots, m\}$ the slope set.

A partition lies in exactly one class:

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\mathrm{special}}(n).$$

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}$$
$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}$$

Proof sketch (Franklin's involution)

Define $F = \alpha$ on $\mathcal{P}_\alpha$, β on $\mathcal{P}_\beta$, and the identity on $\mathcal{P}_{\text{special}}$.

1. $\alpha : \mathcal{P}_\alpha \to \mathcal{P}_\beta$ and $\beta : \mathcal{P}_\beta \to \mathcal{P}_\alpha$ are mutually inverse $\Rightarrow$ F is an involution with $\beta \circ \alpha = \mathrm{id}$, $\alpha \circ \beta = \mathrm{id}$.
2. F preserves n and flips the parity of the number of parts on $\mathcal{P}_\alpha \cup \mathcal{P}_\beta$.
3. Hence α, β pair off even with odd partitions: they cancel in $p_e - p_o$.
4. Only the fixed points $\mathcal{P}_{\text{special}}(n)$ survive: empty unless n is pentagonal, where it is a single staircase $\{k, \ldots, 2k-1\}$ or $\{k+1, \ldots, 2k\}$ of parity $(-1)^k$.

$$\boxed{p_e(n) - p_o(n) = \sum_{S \in \mathcal{P}_{\text{special}}(n)} (-1)^{|S|} = \begin{cases} (-1)^k & n = \frac{3k^2 \pm k}{2} \\ 0 & \text{else.} \end{cases}}$$

Lean formalization (`Aristotle/`) — $\approx$ 1,036 lines, `sorry`-free

Component (file)	Lemmas	Lines	Effort
Definitions (`Defs`)	$\mathcal{P}_\alpha, \mathcal{P}_\beta, \mathcal{P}_{\text{spec}}$, base/slope	90	Low
Helpers (`Helpers`)	run/slope, $b \leq m$, $m \geq 2b$	322	High – manual induction
Disjoint + trichotomy	Lem. 11–12	$\sim$26	Low – `grind`/`simp`
Special = pentagonal	Lem. 13–14	$\sim$230	Med. – arithmetic
Franklin involution	Lem. 16–19	$\sim$120	High – very manual
Bijection & counting	Lem. 20–24	$\sim$135	Med. – `Finset.card`

Easy: definitions, disjointness/trichotomy (one `grind` each), final assembly.
Hard / very manual: reasoning about *slope* (a recursive top-run) and *base/max* under $\cup/\setminus$; the four involution lemmas $\alpha \in \mathcal{P}_\beta$, $\beta \in \mathcal{P}_\alpha$, $\beta \circ \alpha = \mathrm{id}$, $\alpha \circ \beta = \mathrm{id}$.
Library-assisted: cardinality/bijection step rides on mathlib's `Finset`.

Three formalizations of the *same* bijective proof

All three prove Euler's PNT via Franklin's involution; they differ in how much structure is spelled out, the size of the input prompt, and whether the proof closes.

	`Aristotle`	`SlidesInput`	`OnlyWordsInput`
Total lines	**1,350**	**728**	**854**
Files	5	3	3
Status	`sorry-free`	`sorry-free`	1 `sorry`

- `Aristotle` **(1,350):** explicit *three-class* split $\mathcal{P} = \mathcal{P}_\alpha \sqcup \mathcal{P}_\beta \sqcup \mathcal{P}_{\text{spec}}$ with *two separate* maps α, β proven mutually inverse, plus a `FormalPowerSeries` layer linking $p_e - p_o$ to $\prod(1 - X^i)$. Most verbose, fully self-contained.
- `SlidesInput` **(728):** *one* map `franklinMap` (ops A/B via a staircase length `stairLen`); non-fixed points cancel, fixed points characterized as the two staircases $\{k..2k-1\}$, $\{k+1..2k\}$. Same content, tighter packaging.
- `OnlyWordsInput` **(854, was 460):** word-only prompt; rides mathlib's `Finset.sum_involution` and a `topRun` helper. A 2nd run discharged the *involutivity* lemma (`franklinInv_invol`, + ∼400 ln of helpers), leaving *one* `sorry`: the fixed-point sum = `pentagonalCoeff`.

Same mathematics, now a ∼1.85× length spread. Pushing the once-leanest run (OnlyWordsInput, 460 → 854 *over a 2nd run) to close the involution lemma made it longer than the one-shot SlidesInput (728) — discharging* `sorry`*s costs lines, and one still remains.*

Jacobi triple product (plain form)

Jacobi triple product

$$\prod_{m=1}^{\infty}\left(1-q^{2m}\right)\left(1+q^{2m-1}w^{2}\right)\left(1+q^{2m-1}w^{-2}\right) \;=\; \sum_{k=-\infty}^{\infty} q^{k^2}\, w^{2k}.$$

Equivalently, with $z=-w^2$,

$$(q;q)_\infty\,(-z;q)_\infty\,(-q/z;q)_\infty \;=\; \sum_{k\in\mathbb{Z}} z^k q^{k(k-1)/2}.$$

A single bilateral identity that specializes to many classical q-series results.

q-Pochhammer symbols and the q-binomial theorem

Finite / infinite q-Pochhammer symbol:

$$(a;q)_n = \prod_{k=0}^{n-1}(1-aq^k), \qquad (a;q)_\infty = \prod_{k=0}^{\infty}(1-aq^k).$$

Gaussian binomial coefficient: $\displaystyle \binom{n}{k}_q = \frac{(q;q)_n}{(q;q)_k\,(q;q)_{n-k}}.$

q-binomial theorem (finite, then $n \to \infty$)

$$\prod_{k=0}^{n-1}(1+zq^k) = \sum_{k=0}^{n} q^{\binom{k}{2}}\binom{n}{k}_q z^k, \qquad \sum_{n\ge 0}\frac{(a;q)_n}{(q;q)_n}z^n = \frac{(az;q)_\infty}{(z;q)_\infty}.$$

The two Euler identities follow by $a=0$ and $a\to\infty$; the Jacobi triple product is then a short corollary. *(Lean:* `qBinom_finite_thm`, `qBinom_infinite_thm` *— high up-front cost, low marginal.)*

Euler's PNT from the Jacobi triple product

Start from the Jacobi triple product

$$\prod_{m\geq 1}\left(1-q^{2m}\right)\left(1-q^{2m-1}z\right)\left(1-q^{2m-1}z^{-1}\right)=\sum_{k\in\mathbb{Z}}(-1)^k z^k q^{k^2}.$$

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Specialize $q\mapsto x^{3/2},\ z\mapsto x^{1/2}$:

$$\underbrace{\prod_{m\geq 1}(1-x^{3m})(1-x^{3m-1})(1-x^{3m-2})}_{=\prod_{n\geq 1}(1-x^n)} \;=\; \sum_{k\in\mathbb{Z}}(-1)^k\, x^{(3k^2+k)/2}.$$

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The three residue classes mod 3 reassemble the full product $\prod_{n\geq 1}(1-x^n)$, and $(3k^2+k)/2$ ranges over the pentagonal numbers. Hence

$$\boxed{\prod_{n\geq 1}(1-x^n)=\sum_{k\in\mathbb{Z}}(-1)^k x^{k(3k-1)/2}}$$

— exactly Euler's Pentagonal Number Theorem, now an analytic corollary rather than a bijective one.

Formalizing the q-series route: where Aristotle struggled

Unlike the Franklin-involution PNT (**one** Aristotle run each), the Jacobi triple product took ∼6 documented runs to close — two of which produced *no code*, only cost/feasibility analyses.

- **Two planned routes abandoned.** *(i)* Complex analysis / Liouville: needs holomorphicity of infinite products and a Liouville theorem on $\mathbb{C}^*$ — not in mathlib (∼900–1500 lines of new infrastructure). *(ii)* Hermite-style coefficient recurrence ($A_{n,k}$ bookkeeping): ∼1500–2000 lines. Both judged too costly.
- **The crux: annulus → punctured disc.** The Cauchy-product proof first only worked on $\|q\| < \|z\| < 1$; the naive extension hit boundary cases $\|z\| = \|q\|^n$ and stalled at one stubborn `sorry` for a full run.
- **Solution (Aristotle).** Prove the 2nd Euler identity for *all* z (not just $\|z\| < 1$) via the series recurrence $S(z) = (1+z)\,S(qz)$ — this removes the annulus restriction entirely, and the product argument runs directly on $0 < \|z\| < 1$.
- **Infrastructure-heavy core.** Cauchy product over $\mathbb{N} \times \mathbb{N}$ (Fubini + diagonal split), each diagonal evaluated by the Durfee-rectangle identity $S_k = 1/(q;q)_\infty$ — proved by an *analytic contraction recurrence*, not the textbook bijection.

Trajectory: 3 `sorry`s → 1 → 0 across runs; final ∼1,700-line `sorry`-free library — but only after iteration and planning the bijective proof never needed.

LLM usage

Model	Output	Input	Cache write	Cache read
Opus 4.7	4.1 M	61 K	25 M	456 M
Sonnet 4.6	1.02 M	0.6 K	2.76 M	21.4 M
Total (interactive, est.)	**5.1 M**	**62 K**	**28 M**	**477 M**
Aristotle (*est.*)	$\approx$ 55–145 M total — $\sim$9–13 runs total			

- **Interactive cost @ API rates** (standard Opus/Sonnet tiers, *not* the Max plan; recorded + this session, est.): $\approx$ \$1,500 (\$1,465 Opus + \$32 Sonnet; output \$310, cache-read \$685, cache-write \$470), $\sim$15 kWh compute / $\sim$19 kWh with datacenter overhead — $\approx$ driving an EV $\sim$110 km, or a fridge for $\sim$2 weeks.
- **Aristotle (est. — no token logs; from run counts):** OnlyWordsInput 2 runs / 854 ln / 1 `sorry`; SlidesInput 1 run / 728 ln / closed; qSeries+JTP 6 documented runs (2 spent only on feasibility/cost analysis) / 1.7 k ln / closed. Total $\sim$9–13 runs, $\approx$55–145 M tok, $\approx$ \$220–880, $\approx$4.5–22 kWh.
- **Punchline:** the *bijective* PNT closed one-shot where it was guided (SlidesInput, 1 run); the leaner word-only OnlyWordsInput took a 2nd run and still carries 1 `sorry`; the *analytic* JTP route needed $\sim$6 iterative runs — two spent only *planning* — yet the whole Aristotle effort ($\approx$ \$220–880) still came in *below* the interactive exploration around it ($\approx$ \$1,500). Easy proofs are one-shot; hard ones iterate — but generating proofs stays cheaper than talking about them.